

Network Scanning and Reconnaissance Using Nmap

Cyber Security Internship – RD INFRO TECHNOLOGY

Name: YARRAGONDA SWARNA LATHA

Internship Domain: Cyber Security

Task: Task 1 – Network Scanning and Reconnaissance

Tool Used: Nmap 7.98 on Kali Linux

1. Objective

The objective of this task was to perform network reconnaissance using Nmap to identify open ports, running services, and operating system information on the target machine.

2. Introduction

Network reconnaissance is the first phase of a security assessment. It helps identify active hosts, open ports, running services, and operating system details. This information is useful for understanding the security posture of a system.

3. Tools Used

- Kali Linux
 - Nmap 7.98
 - Python HTTP Server
-

4. Target

Target Host: **127.0.0.1 (Localhost)**

5. Command Used

```
sudo nmap -sV -A 127.0.0.1
```

6. Results

The scan successfully detected the following:

- Host was active and reachable.
 - Port **8000/tcp** was open.
 - HTTP service was identified as:
 - SimpleHTTPServer 0.6
 - Python 3.13.12
 - Operating system detection indicated Linux Kernel 5.x/6.x.
-

7. Analysis

The scan successfully identified an active HTTP service running on port 8000. Nmap's service version detection provided information about the running application, while OS detection identified the operating system family. This demonstrates how reconnaissance tools help security professionals understand network services before performing further security assessments.

8. Conclusion

Task 1 was successfully completed using Nmap. The scan identified an open port, detected the running service, and gathered operating system information. This task improved my understanding of network reconnaissance and the use of Nmap in cybersecurity.

9. Screenshots

```
swarna77@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 6.91 seconds  
  
(swarna77@kali)-[~]  
$ sudo nmap -sV -A 127.0.0.1  
[sudo] password for swarna77:  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-01 17:01 +0530  
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 0.00% done  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000066s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
8000/tcp  open  http      SimpleHTTPServer 0.6 (Python 3.13.12)  
|_http-server-header: SimpleHTTP/0.6 Python/3.13.12  
|_http-title: Directory listing for /  
Device type: general purpose  
Running: Linux 2.6.X|5.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6  
OS details: Linux 2.6.32, Linux 5.0 - 6.2  
Network Distance: 0 hops  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 8.60 seconds  
  
(swarna77@kali)-[~]  
$
```

```
swarna77@kali: ~  
Session Actions Edit View Help  
tps://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.65 seconds  
  
(swarna77@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:f0:c2:0e brd ff:ff:ff:ff:ff:ff  
    inet 10.10.10.15/24 brd 10.10.10.255 scope global noprefixroute eth0  
        valid_lft 85927sec preferred_lft 85927sec  
    inet6 fd17:625c:f037:2:55e7:bc8d:3191:d20/64 scope global temporary dynam  
ic  
        valid_lft 85929sec preferred_lft 13929sec  
    inet6 fd17:625c:f037:2:a00:27ff:fef0:c20e/64 scope global dynamic mngttmpa  
ddr noprefixroute  
        valid_lft 85929sec preferred_lft 13929sec  
    inet6 fe80::a00:27ff:fef0:c20e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(swarna77@kali)-[~]  
$
```